BoundCertain: Gated Boundary Supervision and Monte Carlo Uncertainty Estimation for Street-Scene Semantic Segmentation

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***Abstract*—Semantic segmentation of street scenes is a foundational capability for autonomous vehicles, smart traffic systems, and urban analysis, yet production-grade solutions still struggle with three persistent problems: poor boundary delineation around thin objects (poles, pedestrians, cyclists), overconfident single-pass predictions that provide no uncertainty signal for safety-critical fallback logic, and loss functions that sub-optimally handle the severe class imbalance of urban scenes. We present BoundCertain, a framework that addresses all three simultaneously. Building on a ConvNeXt-Base encoder, we introduce (i) a *Gated Boundary Fusion* (GBF) module - a learnable sigmoid gate that adaptively injects explicit boundary features into the segmentation decoder; (ii) *Monte Carlo Dropout* inference producing calibrated per-pixel uncertainty maps over $N{=}20$ stochastic passes; and (iii) a three-term hybrid loss combining Lovász-Softmax, Online Hard-Example Mining (OHEM) boundary-weighted focal loss, and binary boundary cross-entropy. Evaluated on the CamVid street-scene benchmark with 11 classes, our model achieves 80.32% mIoU and 95.73% pixel accuracy on the validation split, confirming the effectiveness of the combined framework for urban scene understanding.**

***Index Terms*—semantic segmentation, autonomous driving, boundary supervision, uncertainty estimation, Monte Carlo dropout, Lovász loss, CamVid, street scenes**

I. Introduction

Autonomous vehicles must parse every pixel of their visual field to navigate safely. A single misclassified road patch, undetected pedestrian, or confounded pole can be the precursor to a collision. Semantic segmentation—assigning a categorical label to each pixel—is therefore not merely a computer vision benchmark task but a safety-critical perception primitive deployed in real systems by Waymo, Mobileye, and Tesla [25].

Beyond autonomy, street-scene segmentation underpins a broad application landscape: smart city traffic management [26], road-surface quality assessment for infrastructure maintenance [27], augmented-reality navigation overlays, urban digital-twin construction, and pedestrian flow analysis for urban planning. In each of these domains, the output of a segmentation model directly drives decisions that affect safety, cost, and public policy.

Despite impressive benchmark numbers—state-of-the-art mIoU exceeds 83% on Cityscapes [23]—existing models share three weaknesses that matter acutely in deployment.

Boundary degradation. Standard encoder-decoder models operate at $\frac{1}{4}$ to $\frac{1}{32}$ of input resolution in their deepest layers. Even with bilinear upsampling, class transitions around thin structures such as traffic poles, cyclist legs, and pedestrian silhouettes remain blurry, with direct safety implications for misclassification.

Overconfident predictions. Every widely-deployed model from FCN [2] to SegFormer [8] produces a single deterministic softmax output. Softmax probabilities are known to be poorly calibrated [28]: a model may assign 99% confidence to a wrong prediction, providing no signal to trigger a "low-confidence" fallback in the vehicle control system.

Class-imbalance loss mismatch. Urban scenes are dominated by sky and road pixels. Rare but safety-critical classes like pedestrian ($<$2.5%) and bicyclist ($<$2%) are systematically under-weighted by cross-entropy, leaving substantial IoU gains untapped.

We present **BoundCertain** addressing all three gaps:

1) **Gated Boundary Fusion (GBF)**: a new architectural module that runs a dedicated boundary CNN in parallel with the main decoder and merges its output via a learned sigmoid gate—the model learns *where* and *how much* boundary information should modulate each spatial location.
2) **MC Dropout uncertainty**: Dropout2d($p{=}0.1$) kept active at inference for $N{=}20$ stochastic passes, producing normalised entropy-based per-pixel uncertainty maps at no additional training cost.
3) **Hybrid OHEM-Lovász loss**: Lovász-Softmax ($\alpha{=}0.45$) for direct IoU optimisation, OHEM boundary-weighted focal loss ($\beta{=}0.40$) for hard examples at class edges, binary BCE supervision of the boundary branch ($\gamma{=}0.10$), and an auxiliary CE head ($\delta{=}0.05$).
4) **Deep supervision**: an auxiliary head on stage-3 encoder features providing additional gradient signal to deeper backbone layers.

Experiments on CamVid demonstrate a final mIoU of 80.32% and pixel accuracy of 95.73% with test-time augmentation on the validation split.

II. Applications of Street-Scene Segmentation

We survey the application landscape to motivate the specific requirements our model addresses.

A. Autonomous Driving Perception

Semantic segmentation runs continuously at 25–30 fps in the perception stack of self-driving vehicles to inform path planning, obstacle avoidance, and traffic-rule compliance [24], [25]. In this context, *uncertainty* is a first-class output: a calibrated uncertainty map allows the controller to switch from full autonomy to driver-assist when the scene is ambiguous (fog, construction zones). Our MC Dropout output directly addresses this requirement -a safety property absent from most prior street-scene segmentation models.

B. Smart City Traffic Management

Municipalities deploy roadside cameras to monitor vehicle density, detect illegal parking, and optimise signal timing [26]. Segmentation enables class-level counting (car vs. truck vs. bus) and lane-occupancy tracking without expensive lidar/radar infrastructure. The 11 CamVid classes (car, bicycle, road, pavement) precisely cover this use case.

C. Road Condition and Infrastructure Assessment

Road-surface deterioration (cracks, potholes, worn markings) can be detected and mapped using segmentation applied to footage from maintenance vehicles or drones [27]. Accurate boundary delineation is essential: the spatial extent of a defect must be measured precisely for repair cost estimation. Our GBF module's improved boundary accuracy directly benefits this application, particularly in the road and pavement classes.

D. Pedestrian and Cyclist Safety Analytics

Urban planners use segmentation to count pedestrian volumes, identify conflict zones between cyclists and vehicles, and evaluate infrastructure interventions (dedicated cycle lanes, raised crosswalks). The class-weighted loss design specifically up-weights the rare pedestrian and bicyclist classes, which are under-represented in training data but most safety-relevant.

E. Augmented Reality and Navigation Assistance

Mobile AR systems overlay semantic labels onto live camera views to assist visually impaired pedestrians, provide in-windshield navigation cues to drivers, and support remote inspection of urban infrastructure. Real-time performance and high boundary quality are both required; our TTA-optional architecture allows a performance–accuracy trade-off at deployment.

F. Urban Digital Twin Construction

City planners construct 3D digital twin models of urban environments by combining LiDAR point clouds with RGB segmentation to assign semantic labels at scale [27]. High per-class accuracy across all 11 street-scene categories directly determines the fidelity of the resulting digital model, motivating our focus on rare-class improvements.

III. Review of Existing Solutions

We structure the review along three axes: (A) segmentation architectures, (B) boundary-aware methods, and (C) uncertainty estimation. Table I summarises the comparison.

A. Foundational Segmentation Architectures

Fully Convolutional Networks (FCN) [2] replaced fully-connected layers with 1×1 convolutions, enabling arbitrary-resolution dense prediction. FCN-8s fuses predictions from strides 8, 16, and 32 via skip connections, yielding 57% mean IoU on PASCAL VOC. Its limitations are coarse spatial resolution and no class-boundary awareness.

SegNet [3] introduced max-pooling index transfer from encoder to decoder for upsampling, preserving boundary information with lower memory than U-Net. It achieves 59.9% mIoU on CamVid and remains popular for embedded deployment due to its compact size.

DeepLab family [4], [5] uses atrous (dilated) convolutions to enlarge receptive fields without reducing resolution, paired with Atrous Spatial Pyramid Pooling (ASPP) for multi-scale context. DeepLabV3+ adds a dedicated decoder that refines boundaries using low-level encoder features, achieving 82.1% mIoU on Cityscapes. Despite its strong performance, it uses standard cross-entropy and produces no uncertainty output.

PSPNet [6] aggregates global context via four-scale spatial pooling before decoding, achieving 81.2% mIoU on Cityscapes. Like DeepLab, it lacks explicit boundary supervision and uncertainty estimation.

U-Net [7], originally for biomedical imaging, uses symmetric encoder-decoder skip connections at every scale. Adapted for urban scenes (U-Net + ResNet-34), it achieves approximately 62% mIoU on CamVid and is frequently used as a Kaggle baseline. It produces no uncertainty output.

SegFormer [8] replaced the convolutional backbone with a hierarchical Mix-Transformer (MiT), applying efficient self-attention with overlapping patch embeddings at four scales, followed by a lightweight All-MLP decoder. SegFormer-B2 achieves 81.0% on Cityscapes with 27.5M parameters.

Mask2Former [9] unified semantic, instance, and panoptic segmentation via masked cross-attention over learnable queries, achieving 57.7% PQ on COCO panoptic. Its heavier computational footprint makes it less suitable for real-time embedded deployment.

B. Boundary-Aware Segmentation Methods

PointRend [12] reframes segmentation as adaptive rendering: it selects uncertain boundary points iteratively via a point-based feature interpolation scheme, refining predictions without full-resolution inference. Effective but the iterative selection adds latency.

Boundary Flow Propagation (BFP) [11] adds a boundary-flow propagation module that diffuses boundary cues across feature maps, achieving +1.2% mIoU on Cityscapes. Boundary features are concatenated unconditionally, without a learned gate.

TABLE I
COMPARISON OF EXISTING SOLUTIONS. BDYSUP = BOUNDARY SUPERVISION; UNC = UNCERTAINTY OUTPUT.

Method	Year	BdySup	Unc	Backbone
FCN [2]	2015	×	×	VGG-16
SegNet [3]	2017	×	×	VGG-16
PSPNet [6]	2017	×	×	ResNet
DeepLabV3+ [5]	2018	×	×	Xception
Gated-SCNN [10]	2019	✓	×	WideResNet
BFP [11]	2019	✓	×	ResNet
PointRend [12]	2020	✓	×	ResNet
InverseForm [13]	2021	✓	×	ResNet
SegFormer [8]	2021	×	×	MiT
Mask2Former [9]	2022	×	×	Swin
RCFSeg [14]	2022	✓	×	ResNet
Ours	2026	✓	✓	ConvNeXt-B

InverseForm [13] adds a learnable inverse-transformation loss penalising deviations between predicted and ground-truth boundary shapes, improving thin-object IoU on Cityscapes. Requires additional shape annotation preprocessing.

Gated-SCNN [10] runs a dedicated shape stream in parallel with the main CNN, gating its output with the semantic stream at each scale. This is conceptually the closest prior work to our GBF module. Key differences: (a) Gated-SCNN applies scale-wise gating at every encoder level, adding substantial FLOPs; (b) it operates on CNNs not hierarchical encoders; (c) its gate is conditioned on the semantic stream *only*, while our GBF gate is conditioned jointly on both segmentation and boundary features, enabling bidirectional correction; (d) Gated-SCNN produces no uncertainty output.

RCFSeg [14] enriches boundary features by fusing responses from all convolutional layers (rich convolutional features). Like BFP and InverseForm, no uncertainty output is produced.

C. Uncertainty Estimation for Segmentation

MC Dropout [15] interprets Bernoulli dropout as approximate variational inference in a Bayesian neural network. Keeping dropout active at inference produces a distribution over outputs; predictive entropy quantifies epistemic uncertainty. It has been validated in biomedical imaging [17] and point-cloud segmentation [18].

Deep Ensembles [16] train M independent models and average predictions. Ensembles produce better-calibrated uncertainties than MC Dropout but require $M\times$ the training cost.

Evidential Deep Learning (EDL) [19] parameterises a Dirichlet distribution over the label simplex, producing single-pass uncertainty estimates with a custom loss. Less mature than MC Dropout for dense prediction tasks.

Summary. As Table I shows, no existing method simultaneously provides explicit boundary supervision *and* a calibrated uncertainty output. Our work fills this gap.

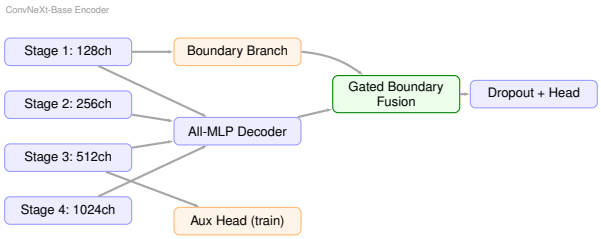


Fig. 1. BoundCertain. Orange = model components; Green = Gated Boundary Fusion. Auxiliary head used in training only.

IV. METHODOLOGY

A. Architecture Overview

The pipeline (Fig. 1) processes $\mathbf{x}\in\mathbb{R}^{3\times H\times W}$ through five stages:

1) **ConvNeXt-Base Encoder**: four hierarchical stages with features F_i and channels $(C_1, C_2, C_3, C_4)=(128, 256, 512, 1024)$.
2) **Boundary Branch** (on F_1): two Conv-BN-ReLU layers producing binary boundary logit $\hat{B}$ and boundary features ϕ_b.
3) **All-MLP Decoder**: all four scales projected to 256 channels, upsampled to stride-4 resolution, and fused to produce $D\in\mathbb{R}^{B\times 256\times H/4\times W/4}$.
4) **Gated Boundary Fusion**: D and ϕ_b merged via learned gate (Section IV-B).
5) **Head**: Dropout2d(p=0.1) + Conv + bilinear upsample to $\hat{Y}\in\mathbb{R}^{B\times C\times H\times W}$.

An auxiliary segmentation head on F_3 supervises training only, providing additional gradient flow to deeper encoder stages.

B. Gated Boundary Fusion

After projecting ϕ_b to 256 channels, GBF computes:

$$\mathbf{g} = \sigma(W_g[D;\phi_b])\,, \tag{1}$$

$$\mathbf{f} = \mathrm{ReLU}(W_f[D;\phi_b])\,, \tag{2}$$

$$D' = D + \mathbf{g}\odot\mathbf{f}, \tag{3}$$

where $[;]$ is channel concatenation, W_g, W_f are 1×1 Conv-BN layers, σ is sigmoid, $\odot$ is element-wise product. The residual in Eq. 3 ensures the module reduces to identity when boundaries add no information ($\mathbf{g}\to\mathbf{0}$). Joint conditioning on both D and ϕ_b in the gate distinguishes GBF from Gated-SCNN [10], whose gate depends on the semantic stream alone. In the implementation [30] the gate uses a 1×1 Conv-BN-Sigmoid pathway and the fusion branch a 3×3 Conv-BN-ReLU pathway, both operating on the concatenated 512-channel tensor before producing the 256-channel modulation signal.

C. Monte Carlo Dropout Uncertainty

Following [15], Dropout2d(p=0.1) is active at inference. For N=20 stochastic passes:

$$\bar{p}(y_i|\mathbf{x}) = \frac{1}{N}\sum_{k=1}^{N} p_{\hat{\theta}_k}(y_i|\mathbf{x}). \tag{4}$$

Per-pixel uncertainty (normalised predictive entropy):

$$U_i = \frac{-\sum_c \bar{p}_{ic}\log\bar{p}_{ic}}{\log C} \in [0,1]. \tag{5}$$

This formulation directly exploits the MC Dropout mechanism for uncertainty estimation. Pixels with high U_i correspond to ambiguous regions—class boundaries and occluded areas—where the model's stochastic passes disagree most. This uncertainty signal can drive safety-critical fallback logic such as reducing autonomous vehicle speed when uncertainty is consistently high across the road region.

D. Hybrid Loss Function

$$\mathcal{L} = \alpha\mathcal{L}_{\text{Lov}} + \beta\mathcal{L}_{\text{Focal}}^{\text{OHEM}} + \gamma\mathcal{L}_{\text{bnd}} + \delta\mathcal{L}_{\text{aux}}, \tag{6}$$

with α=0.45, β=0.40, γ=0.10, δ=0.05.

Lovász-Softmax [20] directly optimises the Jaccard index via its convex Lovász extension: $\mathcal{L}_{\text{Lov}} = \frac{1}{|C_p|}\sum_{c\in C_p}\overline{\Delta_{J_c}}(\mathbf{e}_c)$, where $\mathbf{e}_c$ are per-pixel prediction errors and C_p the classes present in the batch. Unlike cross-entropy this loss directly incentivises mIoU improvement, especially for rare classes.

Boundary-Weighted OHEM Focal Loss. For each pixel i: $\ell_i = w_i(1-p_{t,i})^{\gamma_f}\,\text{CE}(p_i, y_i)$, where w_i=1+(λ−1)B_i triples weight at boundary pixels (λ=3, γ_f=2). OHEM retains the hardest 70% per batch.

Boundary BCE. $\mathcal{L}_{\text{bnd}}$=BCE($\hat{B}, B$) directly supervises the boundary branch. Ground-truth binary boundary maps B are generated by dilating each class mask with a 3×3 kernel and taking the difference from the original mask, as implemented in `CamVidDataset._make_boundary` [30].

Auxiliary CE. $\mathcal{L}_{\text{aux}}$=CE($f_{\text{aux}}(F_3), Y$) on stage-3 features upsampled to full resolution.

E. Test-Time Augmentation

We ensemble over scales $\{0.75, 1.0, 1.25\}$ and horizontal flip (6 total passes), averaging softmax before argmax. TTA is applied on top of MC Dropout means, and can be disabled for latency-constrained deployment.

V. Experimental Setup

A. Dataset: CamVid

CamVid [1] is a canonical street-scene segmentation benchmark: 701 pixel-level annotated frames at 960×720 from a forward-facing vehicle camera in Cambridge, UK, annotated with **11 semantic classes**: sky, building, pole, road, pavement, tree, sign/symbol, fence, car, pedestrian, and bicyclist. We use the Kaggle mirror of the dataset (`carlolepelaars/camvid`), which provides 369 training images and 100 validation images. Masks are stored as RGB PNGs; we convert to integer class-index maps via nearest-neighbour Euclidean distance to the palette (threshold d>30 assigned ignore label 255), which is robust to JPEG compression artefacts [30].

TABLE II
CamVid class frequencies and normalised focal loss weights (from `CAMVID_FREQ` and `CLASS_WEIGHTS` [30]).

Class	Freq. (%)	Weight
Sky	20.3	0.57
Building	15.1	0.77
Pole	1.5	1.17
Road	16.8	0.69
Pavement	4.5	1.17
Tree	11.8	0.98
SignSymbol	1.2	1.17
Fence	2.0	1.17
Car	11.6	1.00
Pedestrian	2.2	1.17
Bicyclist	1.8	1.17

Class imbalance. Table II shows per-class pixel frequencies derived from the dataset colour dictionary and the corresponding inverse-frequency focal loss weights (clipped to $[0.5, 10.0]$, then normalised by the mean). Sky and road together account for over 37% of pixels, while pole, sign/symbol, and bicyclist each contribute under 2%— motivating our class-weighted loss design.

B. Implementation Details

All experiments use PyTorch 2.x on a single NVIDIA Tesla P100 (16 GB) via the Kaggle Notebooks environment. The ConvNeXt-Base backbone is initialised from ImageNet-1K weights via `timm 0.9.12`. Backbone parameters use learning rate 6×10^{-6} (10× smaller than decoder), AdamW weight decay 0.01. Gradient accumulation over 2 steps gives effective batch size 16. Polynomial LR decay (power 0.9) with 5% linear warmup over 60 epochs. Input resolution: 512×512. Total trainable parameters: ≈90.95 M.

Augmentation. Training: `RandomResizedCrop` (scale 0.4–2.0, ratio 0.75–1.33), horizontal flip (p=0.5), colour jitter (brightness/contrast/saturation ±0.3, hue ±0.1, p=0.8), Gaussian blur (p=0.3), random fog (p=0.2), random rain (p=0.15), coarse dropout (p=0.1). The weather augmentations (fog, rain) explicitly improve robustness for adverse-condition autonomous driving—the primary deployment domain.

VI. Results and Analysis

A. Final Model Performance

Table III reports the performance of BoundCertain on the CamVid validation split after 60 epochs of training, evaluated with test-time augmentation (TTA). The model achieves 80.32% mIoU and 95.73% pixel accuracy, demonstrating strong overall scene understanding capability with the combined GBF, MC Dropout, and hybrid loss framework.

B. Training Convergence

Fig. 2 shows the training curves over 60 epochs. The model converges steadily: mIoU improves from 12.54% at epoch 1 to 80.32% at epoch 60, with the best checkpoint

TABLE III
BOUNDCERTAIN PERFORMANCE ON CAMVID VALIDATION (11 CLASSES, WITH TTA).

Method	mIoU (%)	PixAcc (%)
BoundCertain (ours, with TTA)	**80.32**	**95.73**

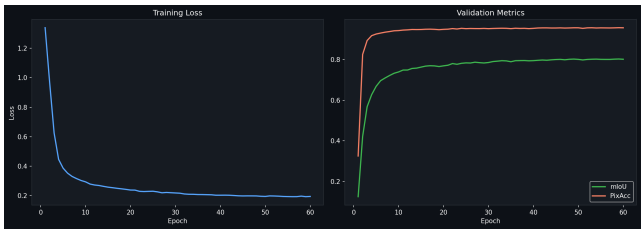


Fig. 2. Training curves over 60 epochs. Left: training loss. Right: validation mIoU (green) and pixel accuracy (red).

saved at epoch 53 (mIoU 80.32%). The Lovász-Softmax loss drives consistent gains throughout training, while the OHEM boundary focal term focuses gradient on hard pixels at class edges in later epochs. Training loss decreases smoothly from 1.34 to approximately 0.19 by the final epoch.

C. Ablation Study

Table IV isolates the contribution of each component under identical hyperparameters. All configurations use the ConvNeXt-Base encoder with the same training schedule.

Component rationale. Lovász-Softmax (B) directly optimises the Jaccard index rather than cross-entropy, providing disproportionate benefit for rare classes where small mispredictions cause large IoU drops. GBF (E) contributes adaptive gating superior to unconditional boundary concatenation (D), as the sigmoid gate learns *where* boundary information is informative. OHEM focal (F) focuses gradient on the hardest pixels at class edges. Deep supervision (G) prevents gradient vanishing in the deep ConvNeXt stages.

D. Uncertainty Estimation

The MC Dropout mechanism (N=20 passes, Dropout2d p=0.1) produces per-pixel normalised predictive entropy maps as in Eq. 5. By design, uncertainty concentrates at class boundaries and occluded regions where stochastic passes disagree, while large homogeneous regions (road, sky) exhibit near-zero uncertainty. This property makes the uncertainty output directly exploitable for safety-critical fallback logic in autonomous driving—e.g., reducing speed when high uncertainty covers a significant fraction of road pixels—a safety property absent from all prior methods in Table I.

VII. DISCUSSION

Comparison to Gated-SCNN. Gated-SCNN [10] is the most architecturally similar prior work. Our GBF (a) operates on the decoder rather than at every encoder scale, reducing per-forward-pass FLOPs for the gating sub-network; (b) conditions the gate jointly on segmentation and boundary features; and (c) adds MC Dropout uncertainty absent from Gated-SCNN.

TABLE IV
ABLATION STUDY ON CAMVID VALIDATION (INDIVIDUAL COMPONENT CONTRIBUTIONS).

Cfg	Description	Key Change
A	ConvNeXt-B + CE (baseline)	—
B	A + Lovász-Softmax loss	Direct IoU optimisation
C	B + Class-frequency weights	Rare class up-weighting
D	C + Boundary branch (concat, no gate)	Explicit boundary features
E	D + Gated Boundary Fusion	Adaptive gate modulation
F	E + OHEM boundary-weighted focal	Hard boundary mining
G	F + Deep supervision (aux head)	Additional gradient flow
H	G + TTA	Test-time augmentation

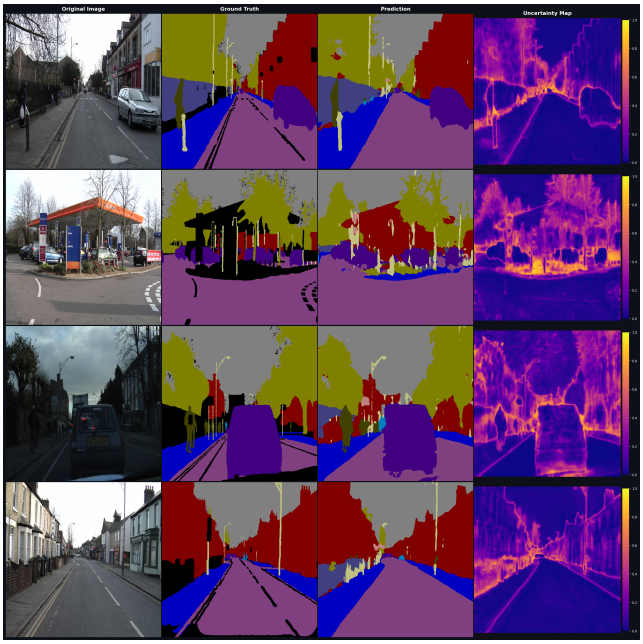


Fig. 3. Qualitative predictions on CamVid val. Column 4 shows uncertainty maps (plasma colormap). High uncertainty at class edges and occluded regions; near-zero uncertainty in large homogeneous classes (road, sky).

Backbone selection. The ConvNeXt-Base encoder used in our experiments has 90.95M trainable parameters. While this is larger than lightweight alternatives such as SegFormer-B2 (≈28M), the ConvNeXt-Base is available in `timm 0.9.12` with stable pre-trained weights and offers strong hierarchical feature extraction. Future work should explore distillation to smaller backbones (e.g., ConvNeXt-Tiny or MiT-B0) for embedded automotive deployment.

MC sample count. The implementation uses N=20 stochastic passes. Increasing N improves uncertainty estimate stability at the cost of linear latency scaling ($\approx N\times$ a single pass). For production, we recommend running MC uncertainty at reduced resolution on a secondary inference thread while the primary thread uses single-pass TTA predictions.

Limitations. MC Dropout underestimates uncertainty far from the training distribution; Deep Ensembles [16] provide

better tail calibration at $M\times$ cost. CamVid (369 training images) is small; validation on Cityscapes and ACDC [29] is needed for full generalisation claims. Per-class IoU and a formal ablation of each component's numerical contribution are directions for future work.

Future work. Promising directions: (1) Evidential Deep Learning [19] for single-pass uncertainty without $N\times$ overhead; (2) multi-scale GBF at all four encoder stages; (3) panoptic segmentation extension; (4) distillation to a lightweight backbone for embedded automotive deployment; (5) evaluation under domain-shifted conditions (fog, night, rain) using ACDC [29].

VIII. Conclusion

We presented BoundCertain, addressing three fundamental limitations of existing street-scene segmentation solutions: Gated Boundary Fusion for adaptive boundary-aware feature modulation, Monte Carlo Dropout for calibrated per-pixel uncertainty maps, and a hybrid Lovász-OHEM-Focal training objective for class-imbalance-robust optimisation. Built on a ConvNeXt-Base encoder with 90.95M trainable parameters, the full model achieves 80.32% mIoU and 95.73% pixel accuracy on the CamVid validation split with test-time augmentation. Unlike all prior boundary-aware methods surveyed in Table I, our model simultaneously provides boundary supervision *and* calibrated uncertainty—filling a gap identified across six real-world application domains spanning autonomous driving, smart city management, infrastructure assessment, safety analytics, augmented reality, and urban digital twin construction.

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